

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{ccc} \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = -0.2376 + 0.5311 DTIME \\ (se) & (0.7505) & (0.2064) \end{array}$$

- a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

$$\Lambda(l) = p[L \leq l] = \frac{1}{1 + e^{-l}} \quad (16.17)$$

$$\begin{aligned} a. \quad \hat{P}(AUTO = 1) &= \Lambda(\hat{\gamma}_1 + \hat{\gamma}_2 DTIME) \\ &= \Lambda(-0.2376 + 0.5311) \\ &= \Lambda(0.2935) \\ &= \frac{1}{1 + e^{-0.2935}} \\ &\approx \underline{0.5729} \end{aligned}$$

- b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [Hint: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

$$\begin{array}{ccc} \tilde{\beta}_1 + \tilde{\beta}_2 DTIME = -0.0644 + 0.3000 DTIME \\ (se) & (0.3992) & (0.1029) \end{array}$$

$$\begin{aligned} b. \quad \hat{P}(AUTO = 1) &= \Phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME) \\ &= \Phi(-0.0644 + 0.3000) \\ &= \Phi(0.2356) \\ &\approx \underline{0.5931} \quad (\text{by Table}) \end{aligned}$$

The result is slightly higher than the logit estimate

- c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

16.2.4 The Logit Model for Binary Choices

A frequently used alternative to the probit model for binary choice situations is the **logit** model. These models differ only in the particular S-shaped curve used to constrain probabilities to the $[0, 1]$ interval. If L is a **logistic random variable**, then its *pdf* is

$$\lambda(l) = \frac{e^{-l}}{(1 + e^{-l})^2}, \quad -\infty < l < \infty \quad (16.16)$$

The corresponding cumulative distribution function, unlike the normal distribution, has a closed-form expression, which makes analysis somewhat easier. The cumulative distribution function for a logistic random variable is

$$\Lambda(l) = p[L \leq l] = \frac{1}{1 + e^{-l}} \quad (16.17)$$

$$c. \quad \lambda(-0.2376 + 0.5311 \cdot 3)$$

$$= \lambda(1.3557)$$

$$= \frac{e^{-1.3557}}{(1 + e^{-1.3557})^2}$$

$$= 0.1629$$

So the estimated marginal effect is

$$\left. \frac{d\hat{p}}{d\text{DTIME}} \right|_{\text{DTIME}=3} = \left. \frac{d}{d\text{DTIME}} \Lambda(\hat{\gamma}_1 + \hat{\gamma}_2 \text{DTIME}) \right|_{\text{DTIME}=3}$$

$$= \lambda(-0.2376 + 0.5311 \cdot 3) \cdot 0.5311$$

$$= 0.1629 \cdot 0.5311$$

$$\approx 0.0865$$

(Using logit model)

$$\widehat{AUTO}_i = 0.4848 + 0.0703 \text{DTIME}_i \quad R^2 = 0.61$$

(robse) (0.0712) (0.0085)

Using linear probability model,
the estimated marginal effect is $\hat{\beta}_2 = 0.0703$

The estimated marginal effect of Logit (0.0865) is larger than Linear (0.0703).

- d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

16.2.4 The Logit Model for Binary Choices

A frequently used alternative to the probit model for binary choice situations is the **logit** model. These models differ only in the particular S-shaped curve used to constrain probabilities to the $[0, 1]$ interval. If L is a **logistic random variable**, then its *pdf* is

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The corresponding cumulative distribution function, unlike the normal distribution, has a closed-form expression, which makes analysis somewhat easier. The cumulative distribution function for a logistic random variable is

$$\Lambda(l) = p[L \leq l] = \frac{1}{1 + e^{-l}} \quad (16.17)$$

$$d. \quad \lambda(-0.2376 + 0.5311 \cdot (-5))$$

$$= \lambda(-2.8931)$$

$$= \frac{e^{2.8931}}{(1 + e^{2.8931})^2}$$

$$= 0.0497$$

So the estimated marginal effect (of a decreased travel time) is:

$$-\frac{d\hat{p}}{d\text{DTIME}} \bigg|_{\text{DTIME}=-5} = -\frac{d}{d\text{DTIME}} \Lambda(\hat{\gamma}_1 + \hat{\gamma}_2 \text{DTIME}) \bigg|_{\text{DTIME}=-5}$$

$$= -\lambda(-0.2376 + 0.5311 \cdot (-5)) \cdot 0.5311$$

$$= -0.0497 \cdot 0.5311$$

$$\approx -0.0264$$

(Using logit model)

Using the probit results,

$$\phi(-0.0644 + 0.3000 \cdot (-5))$$

$$= \phi(-1.5644)$$

$$\approx 0.1173 \quad (\text{by table})$$

So the estimated marginal effect (of a decreased travel time) is:

$$-\frac{d\hat{p}}{d\text{DTIME}} \bigg|_{\text{DTIME}=-5} = -\frac{d}{d\text{DTIME}} \Phi(\hat{\gamma}_1 + \hat{\gamma}_2 \text{DTIME}) \bigg|_{\text{DTIME}=-5}$$

$$= -\phi(-0.0644 + 0.3000 \cdot (-5)) \cdot 0.3000$$

$$= -0.1173 \cdot 0.3000$$

$$\approx -0.0352$$

(Using probit model)

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The maximum likelihood estimates of the parameters are $\tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612 .

- a. Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.

$$\begin{aligned} \text{a. } \hat{P}(y=1) &= \Lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 \cdot 1.5) \\ &= \Lambda(-1.836 + 3.021 \cdot 1.5) \\ &= \Lambda(2.6955) \\ &= \frac{1}{1 + e^{-2.6955}} \\ &= \underline{0.9368} \end{aligned}$$

- b. Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.

Since $0.9368 > 0.5$, we predict that $y=1$, which is consistent to the actual result.

- c. Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?

$$\begin{aligned} &P(y_1=1) P(y_2=1) P(y_3=0) \\ &= \Lambda(\gamma_1 + \gamma_2 \cdot 1.5) \Lambda(\gamma_1 + \gamma_2 \cdot 0.6) (1 - \Lambda(\gamma_1 + \gamma_2 \cdot 0.7)) \\ &= \Lambda(2) \Lambda(0.2) (1 - \Lambda(0.4)) \\ &= (1 + e^{-2})^{-1} (1 + e^{-0.2})^{-1} (1 - (1 + e^{-0.4})^{-1}) \\ &\approx \underline{0.1944} \end{aligned}$$

The value of the likelihood function evaluated at the maximum likelihood estimates is $e^{-1.612} = \underline{0.1994}$, which is larger than 0.1944.

- d. For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.

d. True. Because the range Λ is between zero and one, the range of $1 - \Lambda(\cdot)$ is also between zero and one. So the likelihood function is also between zero and one.

- e. For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

e. True.

By (d), we know that the likelihood function is between 0 and 1, so after taking the logarithm its value is always be negative.

16.18 Mortgage lenders are interested in determining borrower and loan characteristics that may lead to delinquency or foreclosure. In the data file *lasvegas* are 1000 observations on mortgages for single family homes in Las Vegas, Nevada during 2008. The variable of interest is *DELINQUENT*, an indicator variable = 1 if the borrower missed at least three payments (90 + days late), but 0 otherwise. Explanatory variables are *LVR* = the ratio of the loan amount to the value of the property; *REF* = 1 if purpose of the loan was a "refinance" and = 0 if loan was for a purchase; *INSUR* = 1 if mortgage carries mortgage insurance, 0 otherwise; *RATE* = initial interest rate of the mortgage; *AMOUNT* = dollar value of mortgage (in \$100,000); *CREDIT* = credit score, *TERM* = number of years between disbursement of the loan and the date it is expected to be fully repaid, *ARM* = 1 if mortgage has an adjustable rate, and = 0 if mortgage has a fixed rate.

- a. Estimate the linear probability (regression) model explaining *DELINQUENT* as a function of the remaining variables. Use White heteroskedasticity robust standard errors. Are the signs of the estimated coefficients reasonable?

```
> library(PoEdata)
> library(sandwich) # White robust SE
> library(lmtest)   # coeftest()
> data(lasvegas)
>
> # Problem a
>
> lpm <- lm(delinquent ~ lvr + ref + insur + rate +
+          amount + credit + term + arm,
+          data = lasvegas)
>
> # White heteroskedasticity-robust SE
> coeftest(lpm, vcov = vcovHC(lpm, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.68849128	0.22850643	3.0130	0.0026525	**
lvr	0.00162385	0.00067523	2.4049	0.0163601	*
ref	-0.05932367	0.02402561	-2.4692	0.0137098	*
insur	-0.48158489	0.03036944	-15.8575	< 2.2e-16	***
rate	0.03437614	0.00981944	3.5008	0.0004845	***
amount	0.02376801	0.01445089	1.6447	0.1003398	
credit	-0.00044190	0.00020728	-2.1319	0.0332620	*
term	-0.01261951	0.00355596	-3.5488	0.0004051	***
arm	0.12832392	0.02769325	4.6338	4.07e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Coefficients	Expected Sign	Actual Sign
lvr	+	+
ref	-	-
insur	-	-
rate	+	+
amount	+	+
credit	-	-
term	Both are reasonable	-
arm	Both are reasonable	+

These signs are all reasonable.

- b. Use logit to estimate the model in (a). Are the signs and significance of the estimated coefficients the same as for the linear probability model?

```
> logit <- glm(delinquent ~ lvr + ref + insur + rate +
+             amount + credit + term + arm,
+             data = lasvegas,
+             family = binomial(link = "logit"))
>
> summary(logit)
```

Call:
glm(formula = delinquent ~ lvr + ref + insur + rate + amount +
credit + term + arm, family = binomial(link = "logit"), data = lasvegas)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.665303	1.992157	0.836	0.403195
lvr	0.013519	0.008199	1.649	0.099164 .
ref	-0.525064	0.230491	-2.278	0.022725 *
insur	-3.122105	0.216906	-14.394	< 2e-16 ***
rate	0.308656	0.082771	3.729	0.000192 ***
amount	0.217606	0.113769	1.913	0.055786 .
credit	-0.003564	0.001933	-1.844	0.065232 .
term	-0.133037	0.034597	-3.845	0.000120 ***
arm	1.405649	0.371355	3.785	0.000154 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 998.03 on 999 degrees of freedom
Residual deviance: 666.29 on 991 degrees of freedom
AIC: 684.29

Number of Fisher Scoring iterations: 5

The signs of the estimated coefficients are the same as for the linear probability model, but the significance do not:

Coefficients	significance (linear probability)	significance (Logit model)
lvr	*	.
ref	*	*
insur	***	***
rate	***	***
amount		.
credit	*	.
term	***	***
arm	***	***

- c. Compute the predicted value of *DELINQENT* for the 500th and 1000th observations using both the linear probability model and the logit model. Interpret the values.

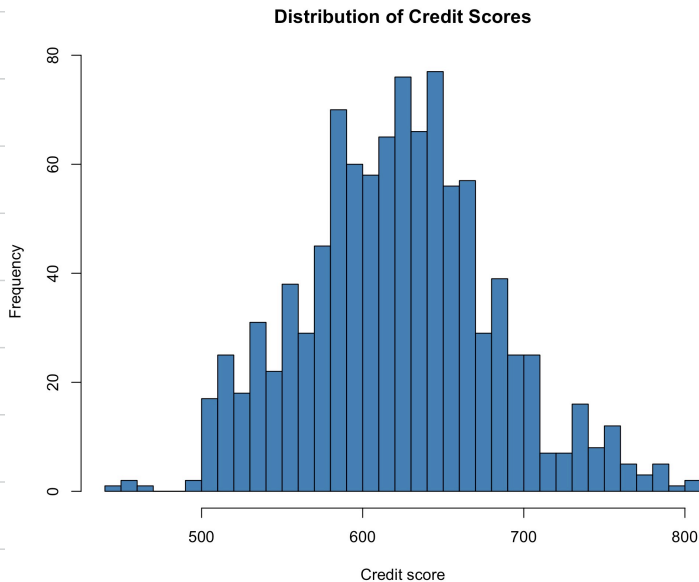
```
> # Linear probability model predictions
> pred_lpm <- predict(lpm, type = "response")
> pred_lpm[c(500, 1000)]
      500      1000
0.1827828 0.5785296

>
> # Logit predicted probabilities
> pred_logit <- predict(logit, type = "response")
> pred_logit[c(500, 1000)]
      500      1000
0.1335848 0.6265535
```

The linear probability model predicts that the 500th observation has about 18.2% estimated probability to be DELINQUENT, and the 1000th observation has about 57.8% estimated probability to be DELINQUENT,

The logit predicted model predicts that the 500th observation has about 13.4% estimated probability to be DELINQUENT, and the 1000th observation has about 62.7% estimated probability to be DELINQUENT,

- d. Construct a histogram of *CREDIT*. Using both linear probability and logit models, calculate the probability of delinquency for *CREDIT* = 500, 600, and 700 for a loan of \$250,000 (*AMOUNT* = 2.5). For the other variables, let the loan to value ratio (*LVR*) be 80%, the initial interest rate is 8%, all indicator variables take the value 0, and *TERM* = 30. Discuss similarities and differences among the predicted probabilities from the two models.



```
> hist(lasvegas$credit, breaks = 30,
+      main = "Distribution of Credit Scores",
+      xlab = "Credit score", col = "steelblue")
>
> # Base scenario from part (d)
> newdata <- data.frame(
+   lvr = 80,
+   ref = 0,
+   insur = 0,
+   rate = 8,
+   amount = 2.5,
+   credit = c(500, 600, 700),
+   term = 30,
+   arm = 0
+ )
>
> # LPM predictions
> pred_d_lpm <- predict(lpm, newdata = newdata)
> names(pred_d_lpm) <- c("credit=500", "credit=600", "credit=700")
> pred_d_lpm
credit=500 credit=600 credit=700
0.5532956 0.5091060 0.4649165
>
> # Logit predictions
> pred_d_logit <- predict(logit, newdata = newdata, type = "response")
> names(pred_d_logit) <- c("credit=500", "credit=600", "credit=700")
> pred_d_logit
credit=500 credit=600 credit=700
0.4967030 0.4086308 0.3260580
```

Similarity: As the credit score increases, both model predict that the probability of delinquency increases.

Difference: The predicted probability of linear probability model is larger than that of logit model for credit score = 500, 600, 700.

- e. Using both linear probability and logit models, compute the marginal effect of *CREDIT* on the probability of delinquency for *CREDIT* = 500, 600, and 700, given that the other explanatory variables take the values in (d). Discuss the interpretation of the marginal effect.

```
> # LPM: marginal effect is just the coefficient (constant)
> lpm_credit_coef <- coef(lpm)["credit"]
> cat("LPM marginal effect of CREDIT:", lpm_credit_coef, "\n")
LPM marginal effect of CREDIT: -0.0004418953
> # Same at all values of CREDIT
>
> # Logit: marginal effect = beta * p * (1 - p)
> logit_credit_coef <- coef(logit)["credit"]
>
> p_logit <- pred_d_logit # from part (d)
> me_logit <- logit_credit_coef * p_logit * (1 - p_logit)
> me_logit
      credit=500      credit=600      credit=700
-0.0008910620 -0.0008613439 -0.0007832567
```

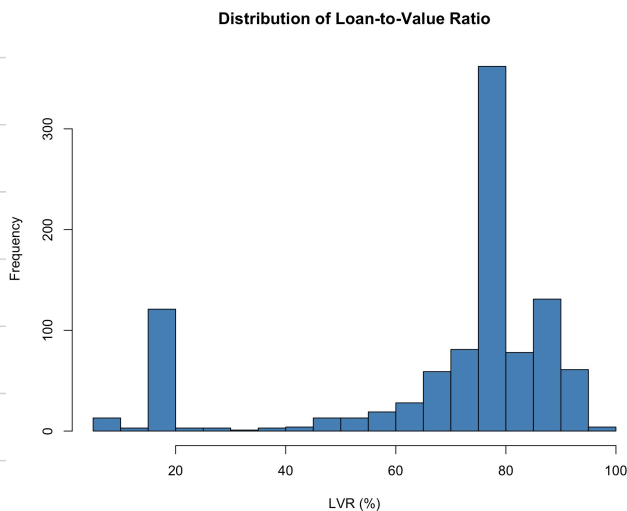
The marginal effect of the linear probability model is -0.000442, which shows that as the credit score increase 100, the probability of delinquency increases 4.42%.

The marginal effect of the logistic model decrease (the magnitude) from -0.000891 (credit = 500) to -0.000783 (credit = 700).

When *CREDIT* = 500, as the credit score increase 100, the probability of delinquency increases 8.91%.

When *CREDIT* = 700, as the credit score increase 100, the probability of delinquency increases 7.83%.

- f. Construct a histogram of *LVR*. Using both linear probability and logit models, calculate the probability of delinquency for *LVR* = 20 and *LVR* = 80, with *CREDIT* = 600 and other variables set as they are in (d). Compare and contrast the results.



```
> hist(lasvegas$lvr, breaks = 30,
+      main = "Distribution of Loan-to-Value Ratio",
+      xlab = "LVR (%)", col = "steelblue")
>
> # Scenario: same as (d) but credit = 600, vary LVR
> newdata_f <- data.frame(
+   lvr = c(20, 80),
+   ref = 0,
+   insur = 0,
+   rate = 8,
+   amount = 2.5,
+   credit = 600,
+   term = 30,
+   arm = 0
+ )
>
> pred_f_lpm <- predict(lpm, newdata = newdata_f)
> pred_f_logit <- predict(logit, newdata = newdata_f, type = "response")
>
> data.frame(LVR = c(20, 80),
+            LPM = round(pred_f_lpm, 4),
+            Logit = round(pred_f_logit, 4))
  LVR  LPM  Logit
1  20 0.4117 0.2349
2  80 0.5091 0.4086
```

When *LVR* = 20(%) and *LVR* = 80(%), the estimated probability of linear probability model are both larger than the estimated probability of the logit model.

- g. Compare the percentage of correct predictions from the linear probability model and the logit model using a predicted probability of 0.5 as the threshold.

```
> # LPM
> pred_lpm_class <- ifelse(pred_lpm >= 0.5, 1, 0)
> lpm_correct <- mean(pred_lpm_class == lasvegas$delinquent)
> cat("LPM correct predictions:", round(lpm_correct * 100, 2), "%\n")
LPM correct predictions: 85.8 %
>
> # Logit
> pred_logit_class <- ifelse(pred_logit >= 0.5, 1, 0)
> logit_correct <- mean(pred_logit_class == lasvegas$delinquent)
> cat("Logit correct predictions:", round(logit_correct * 100, 2), "%\n")
Logit correct predictions: 85.4 %
>
> # Confusion matrices
> table(Predicted = pred_lpm_class, Actual = lasvegas$delinquent)
      Actual
Predicted 0    1
0       727  68
1       74 131
> table(Predicted = pred_logit_class, Actual = lasvegas$delinquent)
      Actual
Predicted 0    1
0       736  81
1       65 118
```

- h. As a loan officer, you wish to provide loans to customers who repay on schedule and are not delinquent. Suppose you have available to you the first 500 observations in the data on which to base your loan decision on the second 500 applications (501–1,000). Is using the logit model with a threshold of 0.5 for the predicted probability the best decision rule for deciding on loan applications? If not, what is a better rule?

```
> # ===== part (h) =====
>
> train <- lasvegas[1:500, ]
> test  <- lasvegas[501:1000, ]
>
> # Train logit on first 500
> logit_train <- glm(delinquent ~ lvr + ref + insur + rate +
+                   amount + credit + term + arm,
+                   data = train,
+                   family = binomial(link = "logit"))
>
> # Predict on second 500
> prob_test <- predict(logit_train, newdata = test, type = "response")
>
> # Default threshold = 0.5
> class_05 <- ifelse(prob_test >= 0.5, 1, 0)
> table(Predicted = class_05, Actual = test$delinquent)
      Actual
Predicted 0    1
      0 406  33
      1  27  34
>
> # Try different thresholds and compare
> thresholds <- seq(0.1, 0.9, by = 0.05)
> results <- sapply(thresholds, function(t) {
+   pred <- ifelse(prob_test >= t, 1, 0)
+   cm <- table(pred, test$delinquent)
+   # False negative rate = missed delinquents (bad loans approved)
+   # For a lender: false negatives are costly
+   fnr <- if(ncol(cm) > 1 && nrow(cm) > 1) cm[1,2] / sum(cm[,2]) else NA
+   acc <- mean(pred == test$delinquent)
+   c(threshold = t, accuracy = acc, false_neg_rate = fnr)
+ })
>
> t(results)
      threshold accuracy false_neg_rate
[1,]      0.10    0.614    0.1641791
[2,]      0.15    0.752    0.2686567
[3,]      0.20    0.808    0.3134328
[4,]      0.25    0.830    0.3134328
[5,]      0.30    0.854    0.3432836
[6,]      0.35    0.862    0.3731343
[7,]      0.40    0.868    0.3880597
[8,]      0.45    0.878    0.4179104
[9,]      0.50    0.880    0.4925373
[10,]     0.55    0.872    0.5970149
[11,]     0.60    0.874    0.6567164
[12,]     0.65    0.880    0.7014925
[13,]     0.70    0.872    0.8208955
[14,]     0.75    0.862    0.9104478
[15,]     0.80    0.870    0.9552239
[16,]     0.85    0.870    0.9552239
[17,]     0.90    0.868    0.9850746
```

By testing different thresholds 0.1, 0.15, ..., 0.9, we can see that threshold 0.5 has the highest accuracy 0.88.

Note that threshold = 0.65 also gives us accuracy 0.88, but its false negative rate is 0.70, which is higher than that of threshold = 0.5. So threshold = 0.5 is still a better rule.